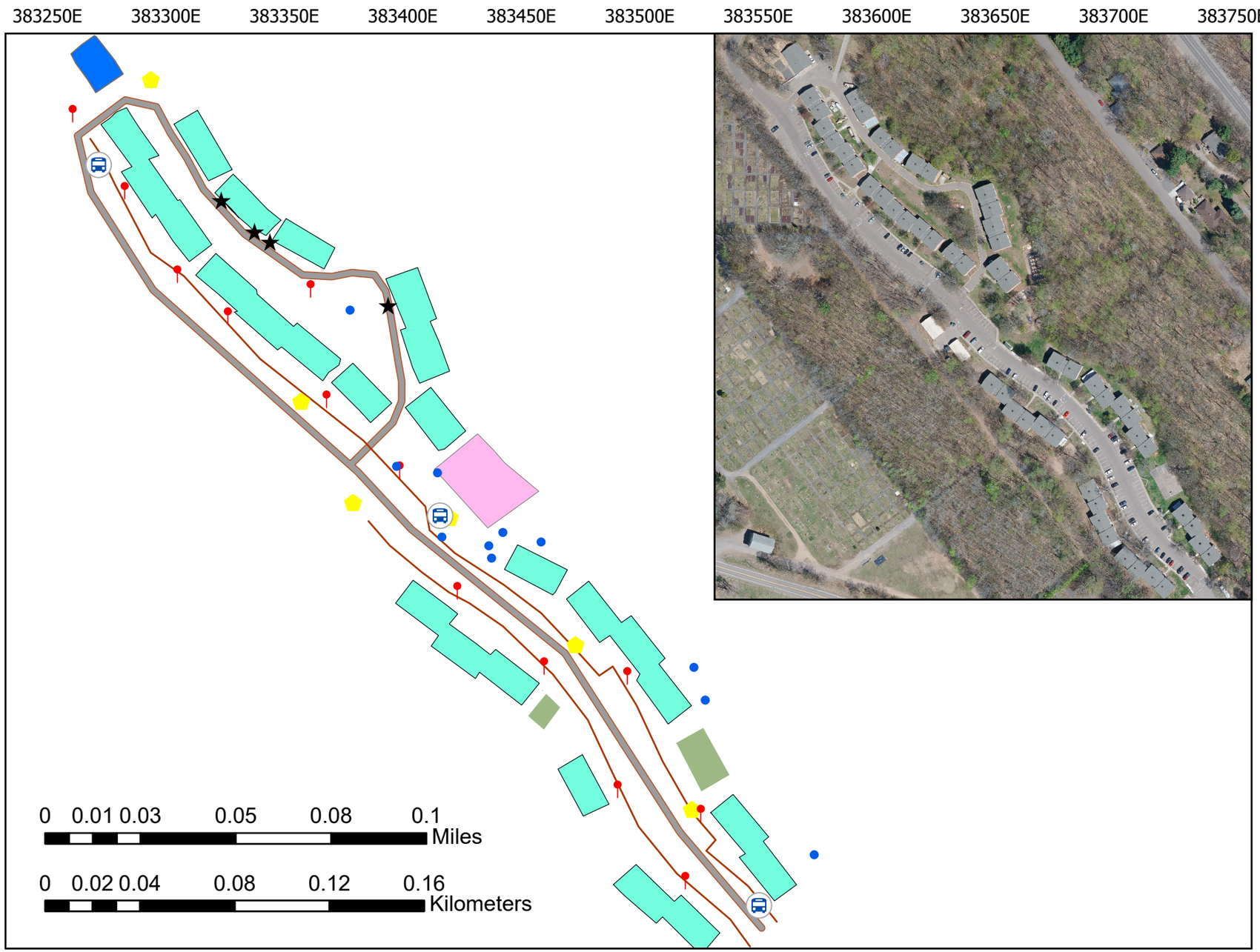
Upper Daniel Heights Asset Management Map



N

Legend

- Benches
- Bus_stops
- Pot_holes
- Street_Lights
- Trash_bins
- Walking_Paths
- Road
- Recreational_areas
- Sports_Facilities
- Pre_School
- Apartments

Author: Wilson Muzavazi
Date: 6 November 2025
Coordinate System: NAD 1983 UTM Zone 16 N
Data Source: Michigan Technological University

Upper Daniel Heights Asset Management, Michigan Technology University

Name: Wilson Muzavazi **Supervisor:** Mike Hyslop

INTRODUCTION:

The Global Positioning System (GPS) is a satellite based navigation system managed by the United States of America. The GPS provides highly reliable information on a user's location, movement and time by transmitting signals that enable receivers to calculate geospatial coordinates (Ahamed, 2010). GPS has developed a critical component of modern businesses, predominantly when combined with Geographic Information Systems (GIS), because it supports accurate mapping and monitoring of physical features on the Earth's surface (El-Rabbany, 2006; Van Sickle, 2021). Spatial accuracy is foundational for informed decision making, especially when managing assets such as campus infrastructure (Ismaeil et al., 2024; Di Giuda et al., 2024).

Effective campus asset management requires up to date and accurate location based data to support maintenance planning, infrastructure monitoring, and safety management (Tripathi et al., 2023). In a university setting, the geospatial enabled GPS field collection has been widely used to map outdoor infrastructure including water lines, electrical systems, buildings, and other facilities because it enhances the quality of spatial inventories and long-term planning processes (Ismaeil et al., 2024; Kang & Tejavanija, 2004).

This project therefore focused on developing a geospatial database of outdoor infrastructure at Upper Daniel Heights, a residential area at Michigan Technological University. GPS data collected through ArcGIS Field Maps and processed in ArcGIS Pro were used to map, visualize, and organize key campus assets with the aim of improving infrastructure management, supporting planning activities, and enhancing campus safety.

Project Objectives:

The main objectives were:

1. To examine how accurate spatial data is when comparing field maps data with (MTU 3inch) Houghton and Hancock drone imagery.
2. To create a geodatabase and map of outdoor assets at Upper Daniel Heights.
3. To support campus facilities management, safety planning, and maintenance.

Research Questions:

1. How accurate is the spatial data collected using Field Maps when compared to the MTU 3-inch Houghton and Hancock drone imagery?
2. What types and distributions of outdoor assets can be effectively mapped and stored in a geodatabase for Upper Daniel Heights?
3. How can the generated geospatial data support campus facilities management, safety planning, and maintenance operations?

Justification of study:

The study was conducted to address the increasing demand for spatial accurate information to implement infrastructure and asset management at Michigan Technological University's Upper Daniel Heights. Traditional asset documentation is often outdated and lacks spatial accuracy. Utilizing the field maps and ArcGIS Pro utility enhances data collection, asset visualization and decision making for campus management.

Significance of study:

The study provides accurate spatial data to support effective asset management, maintenance planning, and safety improvement at Upper Daniel Heights. It demonstrates the value of mobile GIS tools for enhancing infrastructure management and decision-making.

Study Area:

The study area is the Upper Daniel Heights Residential Campus at Michigan Technological University. This area contains student apartments, preschool building, open space recreational area sport facilities and supporting infrastructure which includes roads, trash bins and streetlights.

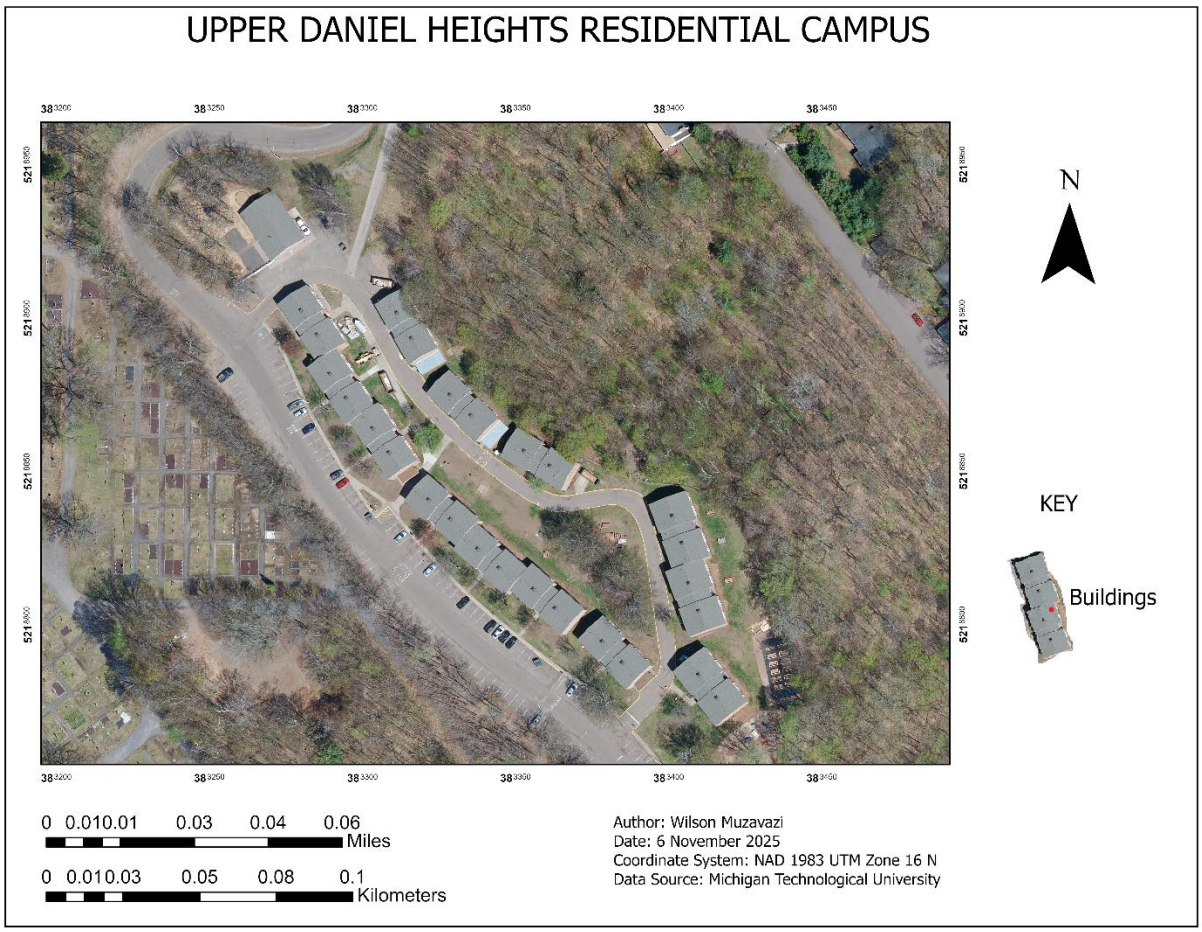


Figure 1 shows the study area map.

METHODOLOGY

Data Collection:

The Mobile application tool ArcGIS field maps was used for the Asset management data collection project. Attribute information was collected, and photos were also attached as part of the data dictionary.

Data dictionary:

A data dictionary was constructed to describe all attribute fields used during the project data collection and post processing. The data dictionary contains field names, categories, data types, units, description, and examples used within ArcGIS Field Maps.

Field Name	Data Type	Description	Examples
Transportation	Text	Type/category of transportation path.	*Road*, *Walking_Path*, *Bike_Path*, etc.
Road_name	Text	Name of the road or path.	Road Name
Surface_Material	Text	Material used on the surface of the road/path.	*Concrete*, *Asphalt*, *Gravel*, *Dirt*, etc.
Width	Double	The width of the transportation feature.	Meters
Pavement_Type	Text	Type of pavement applied to the surface.	*Concrete*, *Asphalt*,
Infrastructure_Areas	Text	Apartments	Type of infrastructure the building belongs to (e.g., Apartments).
Use	Text	Residential Area	Describes the primary function of the building (e.g., residential, commercial).
Condition	Text	Good	Indicates the structural condition of the building at the time of survey (e.g., Good, Fair, Poor).
Asset_Points	Text	Asset category/type	Trash_Bins, Bus_stops, Benches
Condition	Text	Physical condition of the asset	Good, Fair, Poor
Color	Text	Color of asset	Green, Brown
Notes	Text	Additional comments on asset	“Needs repainting”, “Well maintained”
Date_Time	Date/Time	10/25/2025 2:14:13 PM	The date and time when the building feature was collected or recorded.
Creator	Text	Username who created the asset	twmuzava_mtu

Figure 2 Shows the data dictionary.

Data Processing and Map Production:

The collected points, lines, and polygons were inspected to evaluate spatial accuracy. The ArcGIS pro utility was used to correct the datasets using the edit functions such vertices and reshape to accurately correct the positional errors and improve layers alignment. After data cleaning, the recorded features were displayed in ArcGIS Pro for visualization and creating symbology.

Validation of Project:

The Houghton and Hancock 3-inch high-resolution imagery captured in 2018 was used to validate the accuracy of the maps produced for the project. The imagery was used as a basemap to check for alignment of map data collected by the field maps mobile application tool.

RESULTS:

The project produced an accurate Asset Management Utility for infrastructure management. The maps highlight the distribution of fundamental assets at the Upper Daniel Heights Residential Campus.

Preliminary Asset Data Collected Using Field Maps Showing Positional Inconsistencies:

Figure 3 shows the raw data collected using the Field Maps mobile application demonstrated inconsistencies associated with using smartphones for spatial data collection. When the collected polygons were overlaid with the high resolution 3-inch imagery of Houghton and Hancock in ArcGIS Pro, noticeable misalignments were observed. These discrepancies can be attributed to multipath errors and the inherent positional inaccuracies of smartphone GPS sensors, which affect the precision of location data captured in the field.

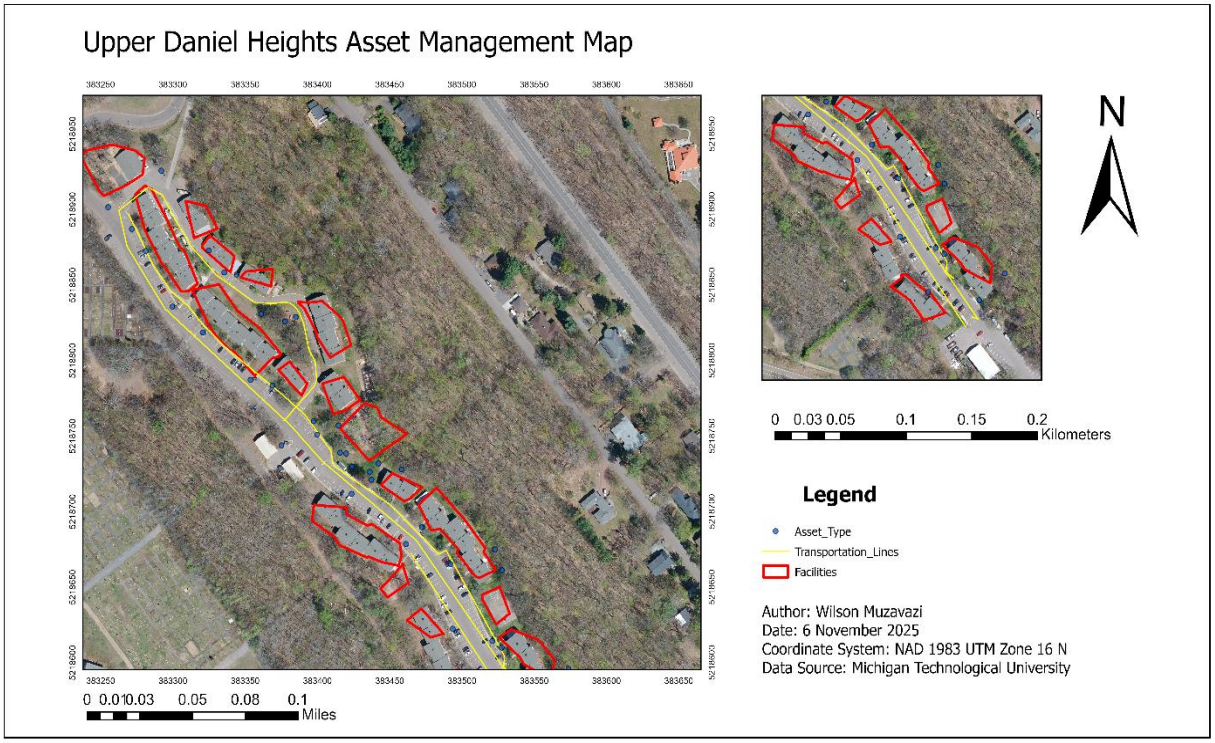


Figure 3 shows Raw data collected using Field Maps.

Corrected Asset Map After Data Cleaning Showing Alignment with 3-Inch Houghton Imagery

Figure 4 shows the final map produced after data cleaning for the Asset Management Project demonstrated a significant improvement in spatial accuracy and consistency. The polygon data layers showed strong alignment with the Houghton 3-inch high-resolution imagery in ArcGIS Pro, indicating effective correction of positional errors identified in the raw dataset. This improvement highlights the importance of post processing and data validation procedures in enhancing the reliability of spatial data collected using mobile devices asset management applications.

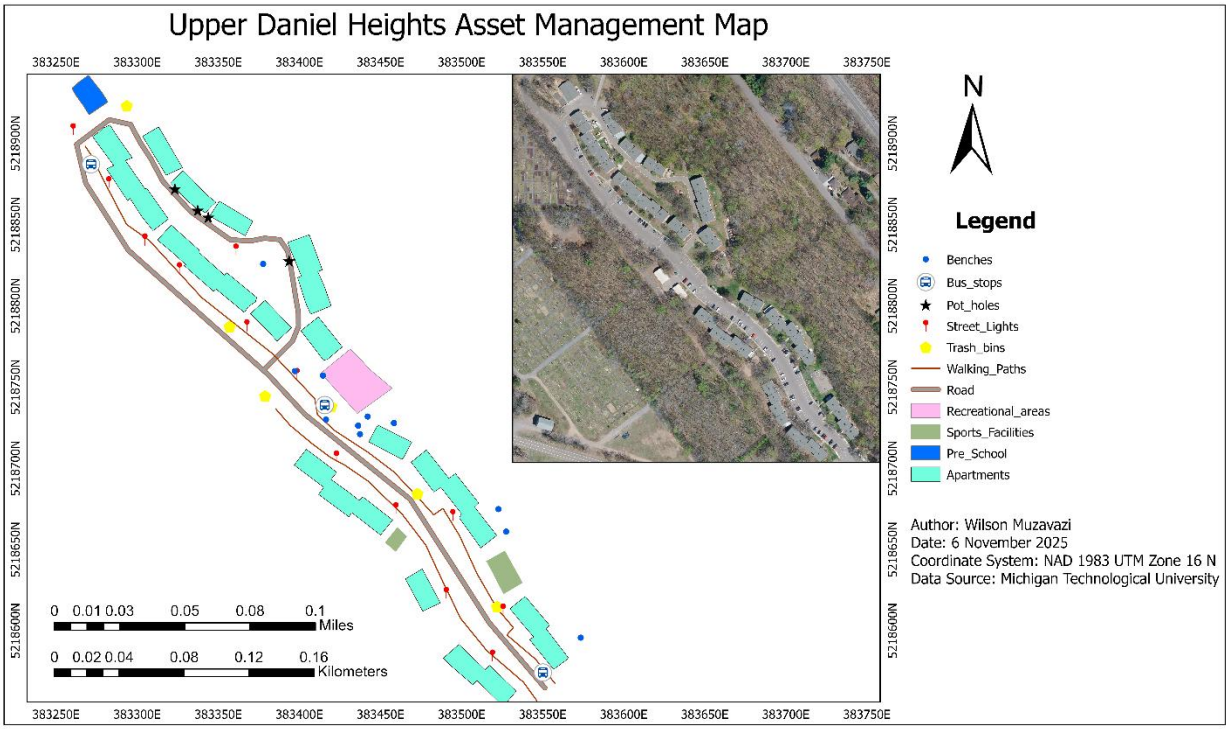


Figure 4 shows the Corrected Map.

DISCUSSIONS:

The asset mapping project at Upper Daniel Heights demonstrated the effectiveness of mobile GIS tools, such as Field Maps, in collecting and managing spatial data for infrastructure management. Although the initial raw dataset revealed positional inaccuracies due to smartphone GPS limitations and multipath errors, post-processing

and data cleaning significantly improved spatial precision. The final dataset showed strong alignment with the high-resolution 3-inch Houghton imagery, confirming the reliability of the corrected data for decision-making applications.

The spatial distribution analysis of mapped features provided valuable insights into the layout and accessibility of key assets. Streetlights were concentrated along major pedestrian routes, contributing to safety and visibility, while recreational and sports facilities were strategically positioned near residential zones to promote accessibility. The uniform distribution of trash bins and benches around communal spaces indicates efficient planning for convenience and cleanliness.

Integrating these datasets into Michigan Technological University's asset management framework offers substantial benefits for maintenance planning, energy management, and sustainability efforts.

Limitation

The study faced numerous limitations that impacted data collection and accuracy. Certain records comprise of data entry errors during field mapping. The Mobile field map faced rapid battery drainage during data collection. My Mint mobile service provider had unstable internet connectivity sometimes interrupted data uploads. Also, the GPS accuracy from the mobile utility produced a lot of spatial misalignments, emphasizing the need for high precision equipment in future studies such as drones. GPS is restricted for indoor building management, corresponding to Kang & Tejavanija (2004), because satellite signals don't penetrate well. The MTU 3-inch Houghton and Hancock imagery requires updating, as the dataset continues to display an apartment building that was destroyed by fire in 2019. The presence of outdated features in the imagery introduces temporal inaccuracies that negatively affect the reliability of my spatial analysis.

If I had to redo the project, what would I have done?

If I were to redo this project, I would implement several methodological improvements to increase the accuracy and reliability of the spatial data collected. First, I would conduct

formal GNSS planning prior to fieldwork to assess satellite visibility, predicted dilution of precision (DOP), and potential obstructions in the study area. Proper GNSS planning helps identify optimal time windows for data collection and reduces positional error, particularly in environments with canopy or building interference (El-Rabbany, 2006; Van Sickle, 2021). Based on the planning results, I would schedule all field activities during periods of high satellite availability and low DOP to ensure better signal geometry and more consistent accuracy (Hofmann-Wellenhof et al., 2008).

CONCLUSIONS

The project successfully demonstrated the use of mobile GIS tools for efficient asset mapping and infrastructure data management at Upper Daniel Heights Residential Campus. The final maps present comprehensive information into the spatial distribution of key assets such as streetlights, recreational areas, and sports facilities ensuring improved planning and maintenance scheduling. Although there are slight limitations associated with GPS accuracy and utility performance, the project shows the capability of Field Maps and ArcGIS Pro for campus asset management.

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